Kantipur City College

Department of Computer Engineering

Semester IV/ Probability and Statistics

Lab Sheet No.8

Title: Binomial Probability Distribution using EXCEL.

Objectives:

i) To calculate Binomial probabilities.

ii) To fit Binomial Distribution

Theory:

Probability model of binomial distribution

Let the random variable X follows binomial distribution that is satisfying above condition in which the probability of success for each trial is p and failure q, then the probability of X successes out of n independent trials is given by

P(X=x) =p(x) = n_{C_x} p ^{x} q $^{(n-x)}$ where x=0,1,2…….n

= 0 otherwise

Symbolically, X ˜ B(n,p)

Fitting of Binomial distribution:

If n independent trials are repeated N times and satisfying the condition of Binomial distribution, then the theoretical or expected frequencies of x success is given by

f(x)= N n_{C_x} p ^{x} q $^{(n-x)}$ where x=0,1,2…….n

If p is not given we evaluate the value of mean with np to find the values of p and q.

i.e $\bar{X}$ = np and $\bar{X}$=$\sum \frac{fx}{N}$

Lab Work:

1. For a Binomial distribution with n=10 and p=0.32 find

i) P(X=5) ii) P(X<5) iv)P(X<=5) v)P(X>=6) vi) P(X>6) vii) P(3<X>8) vii) P(3<=X>=8)

Using excel:

	A	B	C	D	E	F	G	H	I
1	Lab work 1:								
2									
3			Formula						
4	n	10							
5	p	0.32							
6	q	0.68	=1-B2						
7	P(x=5)	0.122941	=BINOM.DIST(5,B4,B5,0)						
8	P(x<=5)	0.936285	=BINOMDIST(5,B4,B5,1)						
9	P(x<5)	0.813345	=BINOMDIST(4,B4,B5,4)						
10	P(x>=6)	0.063715	=1-BINOMDIST(5,B4,B5,1)						
11	P(x>6)	0.015503	=1-BINOMDIST(6,B4,B5,1)						
12	P(3<x<8)	0.401824	=BINOMDIST(7,B4,B5,1)-BINOM.DIST(3,B4,B5,1)						
13	P(3<=x<=8)	0.668471	=BINOMDIST(8,B4,B5,1)-BINOMDIST(2,B4,B5,1)						
14									

Lab work 2;

Fit a binomial distribution to the following data:

X	0	1	2	3	4
f	28	62	46	10	4

Using Excel:

16	Lab work 2:					
17	X	f	expectd frequency f(x)=Np(x)	Formula	Rounded expected frequency	Formula
18	0	28	29.62962963	=B26*BINOMDIST(A18,B27,B29,0)	30	=ROUND(C18,0)
19	1	62	59.25925926		59	
20	2	46	44.44444444		44	
21	3	10	14.81481481		15	
22	4	4	1.851851852		2	
23	Total	150			150	
24						
25						
26	N	150				
27	n	4				
28	np	1.333333	=SUMPRODUCT(B18:B22,A18:A22)/B23			
29	p	0.333333	=B28/B27			
30	q	0.666667	=1-B29			
31						

Assignments:

Fit Binomial distribution and find i) P(X <3)

i) P(X <3)   ii) P(X <=3)   iii) P(X >3)   iv) P(X >4)   v) P(4<X <7)

X	0	1	2	3	4	5	6	7	8
f	20	100	190	280	350	300	220	120	3